Project 1: Clinical Decision Support System with RAG and KG

Concept

Develop a system that augments a Large Language Model (LLM) with a comprehensive knowledge base of medical literature, clinical guidelines, and anonymized Electronic Health Records (EHRs). The system should assist clinicians in diagnosis, treatment planning, and understanding rare conditions by providing evidence-based, contextually relevant information.

Healthcare Value:

- **Improved Diagnosis:** Help clinicians with difficult or rare diagnoses by rapidly sifting through vast medical literature and patient histories.
- **Personalized Treatment:** Support tailored treatment plans based on a patient's unique profile, medical history, and current best practices.
- **Reduced Information Overload:** Provide concise, evidence-backed answers, saving clinicians time.

Key Features for Demo:

- **Contextual Question Answering:** Ask questions like, "What are the common comorbidities for Type 2 Diabetes in patients over 60 with hypertension, and what are the recommended first-line treatments based on the latest guidelines?"
- **Source Citation and Explainability:** The RAG system must cite the source documents (research papers, clinical guidelines, EHR notes) for its answers, demonstrating transparency and trustworthiness.
- **"What If" Scenarios:** Allow clinicians to explore different treatment pathways or diagnostic hypotheses by asking "What if we consider X treatment given Y patient condition?"
- **Pattern Discovery (RAG-enhanced):** Identify subtle relationships between medications, side effects, and patient demographics across a large corpus of notes that might not be explicitly stated in a single document. For example, discover an emerging trend of a particular adverse drug reaction (ADR) associated with a new combination of drugs, even if individual studies only mention single drug ADRs.

Demonstrate following capabilities:

1. Identifying Potential Drug-Induced Symptoms

Question: “Identify all patients who reported symptoms like 'dizziness' or 'fatigue' in their free-text clinical notes, and who are currently on 'Lisinopril'. For these patients, list their initial diagnoses and the full content of the note where the symptom was mentioned, so a clinician can review if it's a potential side effect.”

Suggested approach: Hybrid: Vector DB for Symptom Extraction + Graph for Medication Link)

Hybrid Benefit:

- *Vector DB (RAG):* Would be used to semantically search the unstructured clinical_notes.content for mentions of "dizziness" or "fatigue" (and synonyms/related phrases), allowing flexible natural language queries for symptoms.
- *Graph DB:* Once patients with relevant symptoms are identified, the Graph DB would efficiently link these patients to their medications_on_record (specifically "Lisinopril") and initial_diagnoses, providing structured context.
- *Nested/Discovery:* Discovers potential drug-symptom correlations by connecting unstructured symptom data (from notes) to structured medication data. It allows "discovery" of potential adverse drug reactions.

2: Tracking Symptom Evolution in a Specific Patient Cohort

Question: “For all male patients over 60 years old who have an initial diagnosis of 'Hypertension', trace their clinical_notes chronologically. Are there any patients in this group who initially reported 'dizziness' but later reported 'headaches' after being prescribed 'Lisinopril'? If so, list their patient ID and the dates of these specific symptom mentions.”

Suggested approach:
Graph DB - Nested/Discovery & Temporal Analysis)

Graph DB Benefit: Efficiently filters patients by multiple structured attributes (gender, age, initial_diagnoses). It then allows traversal through nested clinical_notes based on date to detect a sequence or evolution of symptoms.

Nested/Discovery: Demonstrates navigating deeply nested data (clinical_notes array, then date and content within each note) and discovering temporal patterns in symptom presentation related to a specific diagnosis and medication.

3. Finding Patients with Co-occurring Conditions and Specific Complaint Patterns

Question: "Find patients who have both 'Type 2 Diabetes' and 'Hypertension' as initial diagnoses. Among these, identify anyone whose chief complaint in *any* clinical note semantically matches 'unusual thirst and frequent urination'. For such patients, list their medications_on_record and the date of the relevant note."

Suggestd Approach
Hybrid: Vector DB for Chief Complaint + Graph for Multi-attribute Filtering

Hybrid Benefit:

- Vector DB (RAG): Semantic search on the unstructured clinical_notes.content (specifically extracting the Chief complaint: part) for the phrase "unusual thirst and frequent urination," finding patients even if the exact wording varies.
- Graph DB: Efficiently identifies patients based on the structured initial_diagnoses ('Type 2 Diabetes', 'Hypertension') and then links to the medications_on_record for contextual information.

Nested/Discovery
Combines precise structured filtering with flexible unstructured content search to identify a highly specific patient cohort and relevant clinical context for decision support.

4. Identifying Patients with Recurring Symptoms on Specific Medications "Which medications on record are most frequently associated with patients reporting 'fatigue' in their clinical notes on *multiple distinct dates*? For the top 3 such medications, list the patient IDs and dates of notes where 'fatigue' was reported, illustrating recurring symptom patterns."

Suggested Approach:
Graph DB - Nested & Aggregation

Graph DB Benefit: This requires iterating through patients, then their notes, identifying symptom mentions, linking back to medications, and then performing aggregation (counting distinct symptom mentions per patient per medication) to find recurring patterns.

Nested/Discovery: Explores recurring patterns of symptoms within nested clinical_notes and correlates them with medications_on_record, allowing the "discovery" of potential drug-symptom associations based on frequency.

5. Providing Contextual Information for a Patient with New Symptoms

A patient (e.g., Richard Frank, P002) presents with a new complaint of 'sudden blurry vision'
Help the doctor by

a) providing summary history of Richard Frank, any earlier diagnosis such as high BP/diabetes

b) searching clinical notes for blurry vision or semantically similar terms and what other patients with this symptom were diagnosed with. Print the name, symptom, diagnosis and date diagnosed for those patients

Data Sources:

Generate synthetic dataset that contains the patterns to meaningfully answer the questions/address the scenarios below

Tip: copy paste the entire text above into your favourite AI chatbot such as Claude4 and ask it to create python script to generate the dataset that exhibits the patterns needed to answer above questions. Ask it to also validate the patterns in the script.

A reference dataset is provided for you in the project1-dataset folder

The provided **synthetic dataset** contains:

100 patient records and saved to synthetic_patient_ehr.json

30 medical documents and saved to synthetic_medical_knowledge.json

Patterns are verified in the data:

--- Subtle Drug Interaction Pattern ---

Patient P000 (Christopher Chan) on DrugA and DrugB shows 'unusual fatigue' in note from 2024-01-09.

--- T2D + Hypertension Comorbidity Pattern ---

Patient P000 (Christopher Chan) with T2D and Hypertension shows comorbidity mention in note from 2024-09-14.